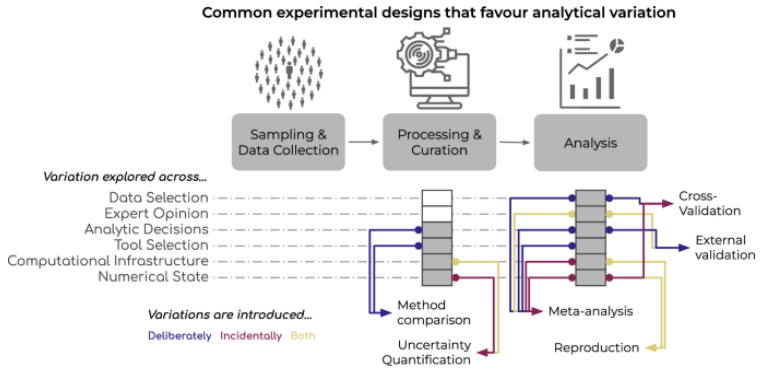


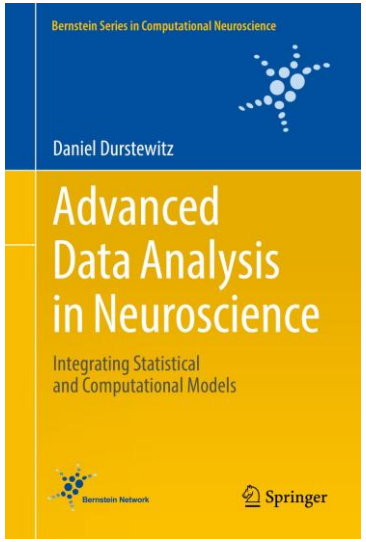
Processing and interpretation of neuroscience data:

Module 1 – Data overview and single cell sequencing

Dr. Kaja Moczulska & Dr. Łukasz Piszczek

Univ.-Prof. Dr. Wulf Haubensak
Department of Neuronal Cell Biology

Center For Brain Research (Zentrum für Hirnforschung)



Overview

Resources needed:

- Laptop
- Internet access
- Materials:
  - VM machine - Docker
  - https://github.com/HugoMalagon/NeuroData_860.053-MUW

„Exam“: Assessment via email after each class

07.10
- Introduction to R, basics
- Visual analytics

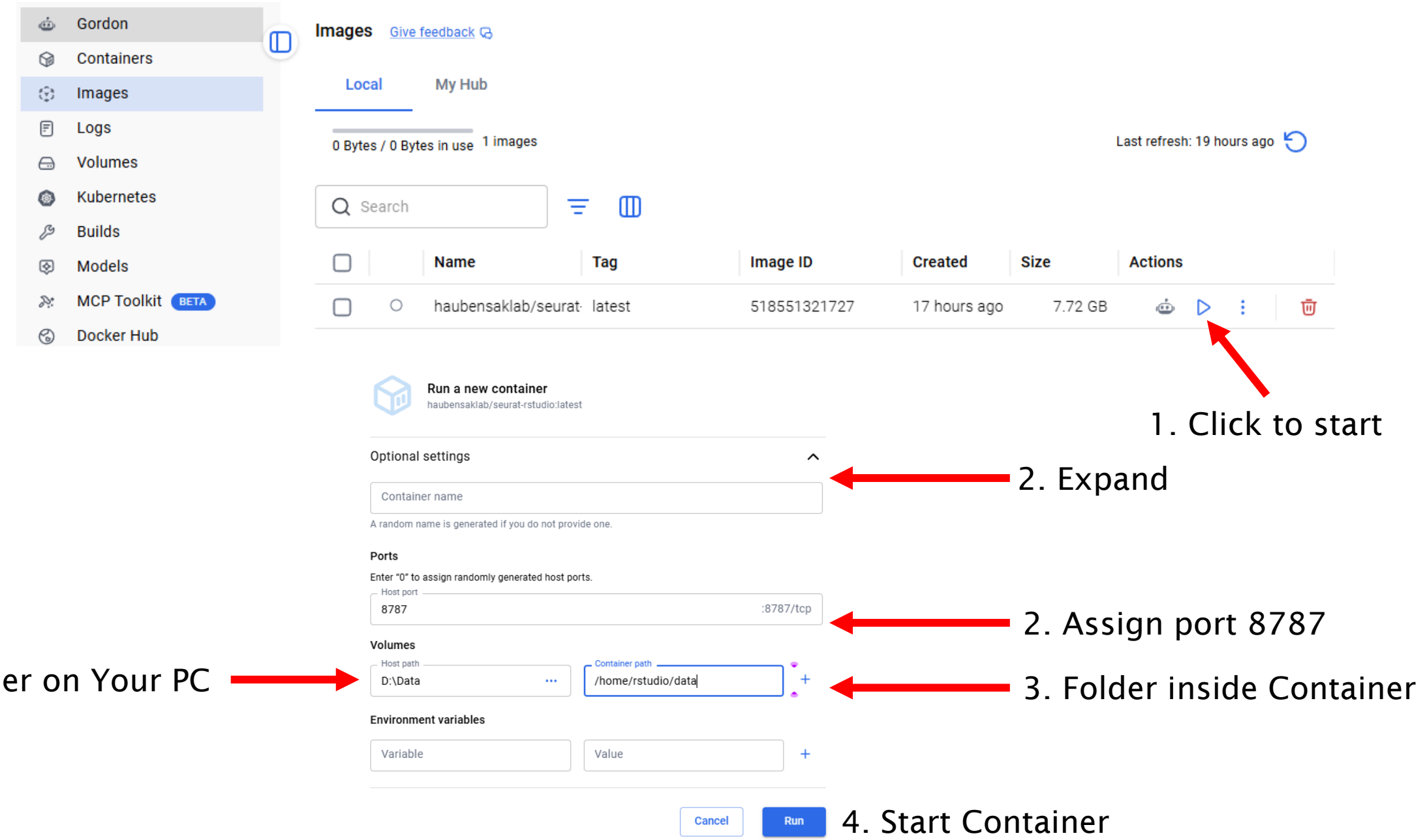
14.10
- dimensional reduction: PCA, UMAP
- Normalization/scaling
- clustering: k-means, knn

21.10
- Intro to Seurat
- Single cell RNA seq
- Dataset merging and preprocessing

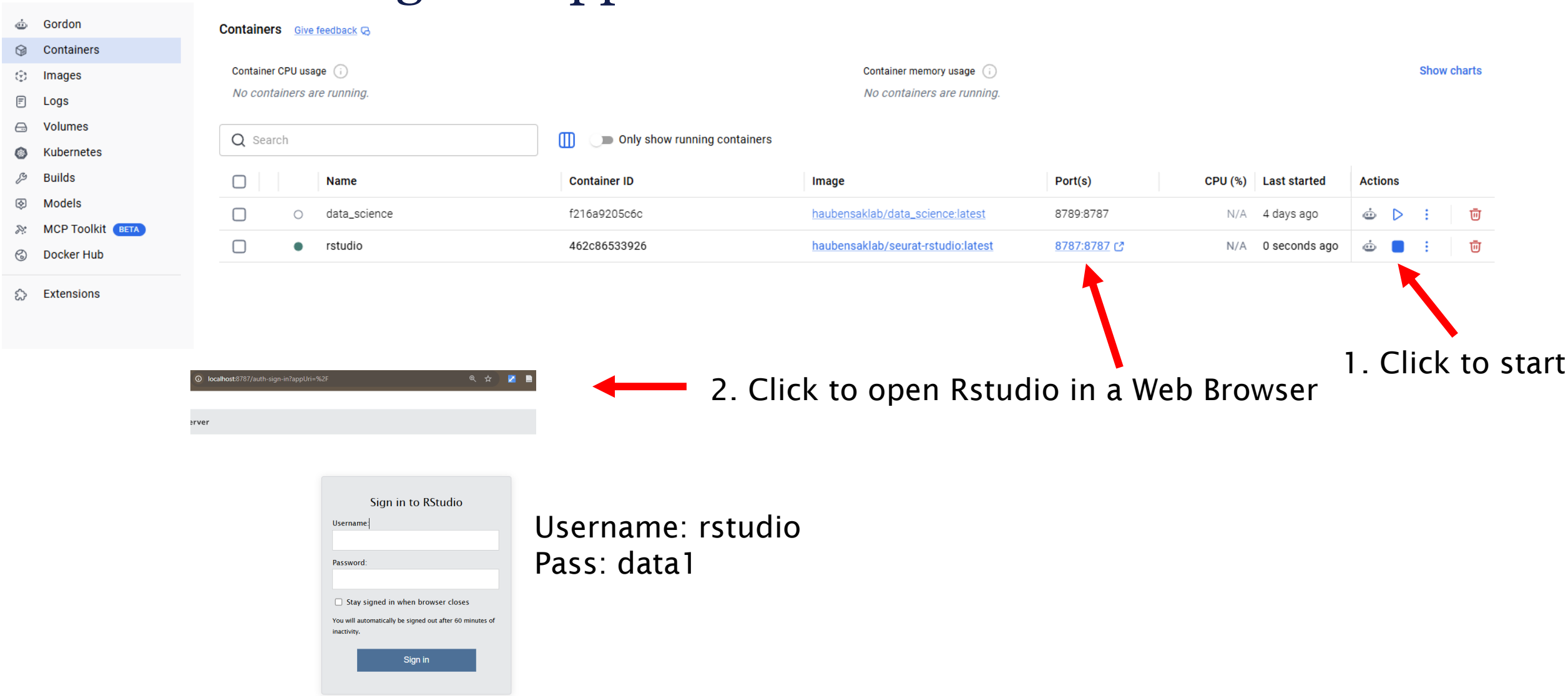
04.11
- clustering in Seurat
- Finding cell types

11.11
- Joining experiments
- DEG interpretation

Docker


Gordon
Containers
Images
Logs
Volumes
Kubernetes
Builds
Models
MCP Toolkit BETA
Docker Hub
Images
Give feedback
Local
My Hub
0 Bytes / 0 Bytes in use
1 images
Last refresh: 19 hours ago
Search
Name
Tag
Image ID
Created
Size
Actions
haubensaklab/seurat·
latest
518551321727
17 hours ago
7.72 GB
1. Click to start
Run a new container
haubensaklab/seurat-rstudio:latest
Optional settings
2. Expand
Container name
A random name is generated if you do not provide one.
Ports
Enter "0" to assign randomly generated host ports.
Host port
8787
:8787/tcp
2. Assign port 8787
Volumes
3. Folder on Your PC
Host path
D:\Data
Container path
/home/rstudio/data
3. Folder inside Container
Environment variables
Variable
Value
Cancel
Run
4. Start Container

Docker – running or stopped container



Username: rstudio
Pass: data1

Overview

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- Internet access
- Materials:
  - VM machine - Docker
  - https://github.com/HugoMalagon/NeuroData_860.053-MUW

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07.10
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Day 3

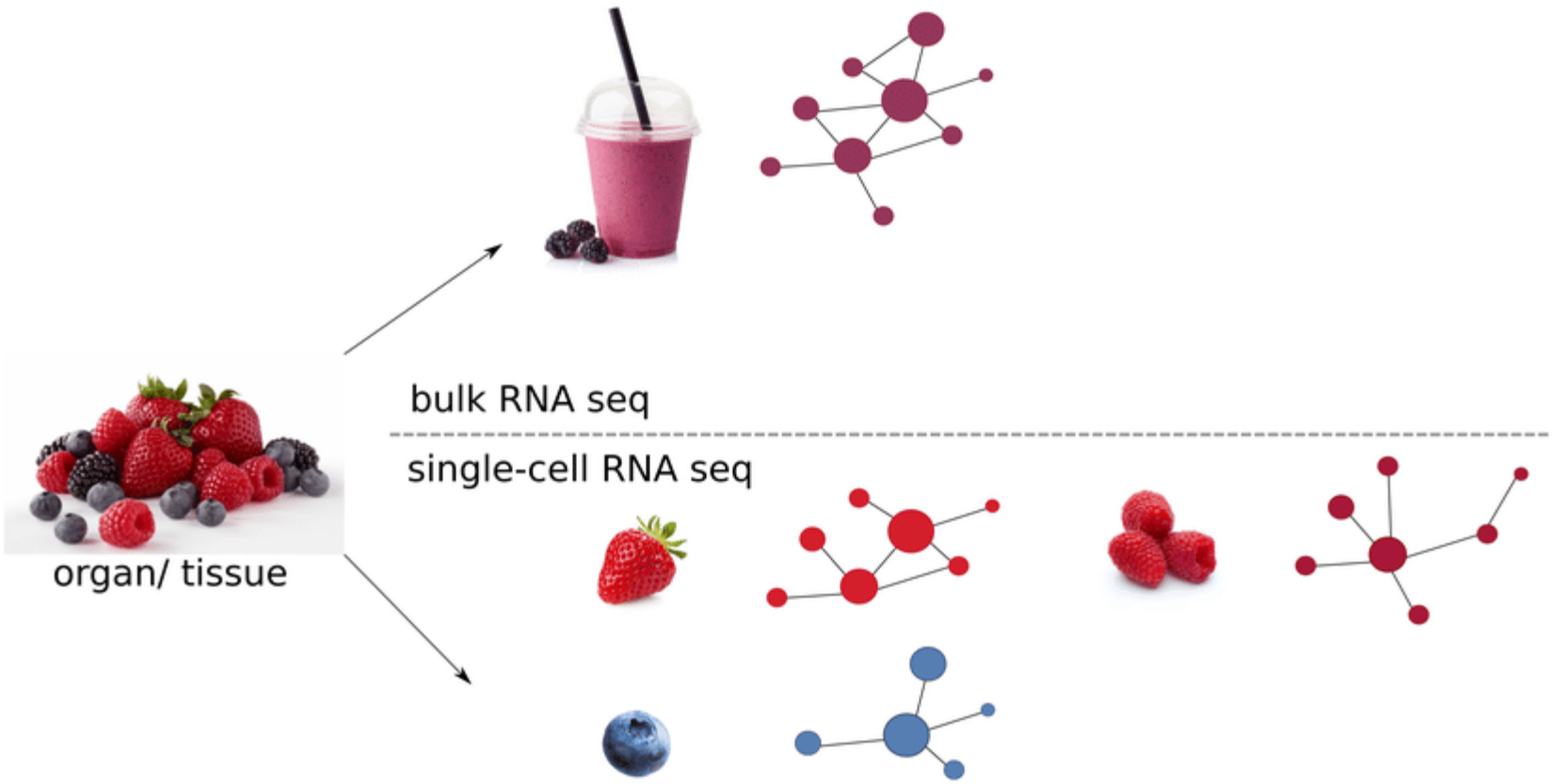
Intro to Seurat

Preprocessing of scRNAseq

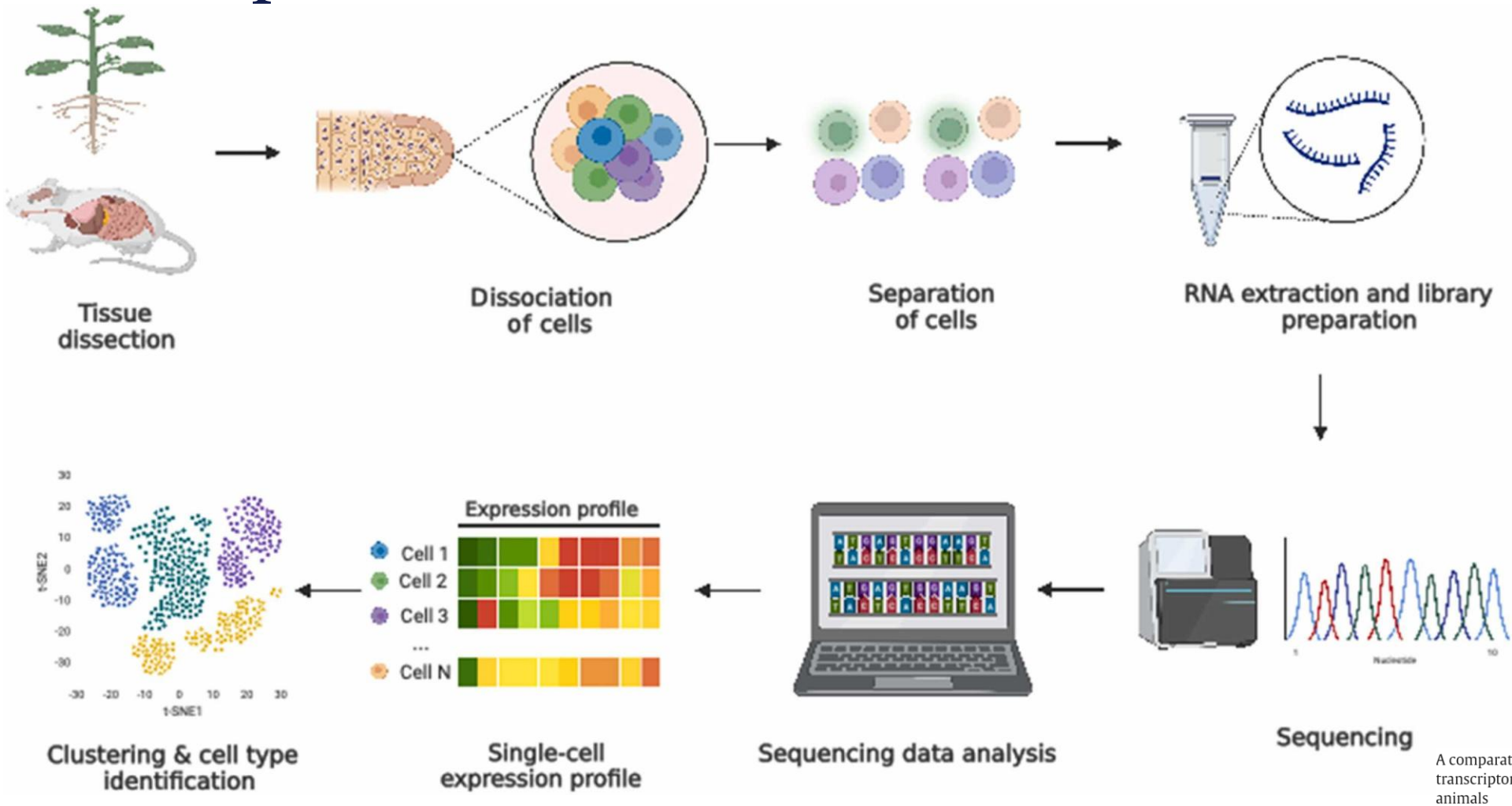
How to measure gene expression?

Method	Throughput	Quantitative	Sensitivity	Main Limitation
Northern Blotting	Low	Semi-quant	Moderate	Low throughput, laborious
qPCR	Low-Medium	Quantitative	High	Target-specific, limited multiplexing
Microarray	High	Quantitative	Moderate	Lower sensitivity than RNA-Seq
RNA-Seq	Very High	Quantitative	Very High	Expensive, computational load
In Situ Hybridization	Low	Qualitative	Moderate	Labor-intensive
SAGE	High	Quantitative	High	Largely replaced by RNA-Seq
Reporter Assays	Variable	Quantitative	Variable	Requires reporter construct
Digital PCR (dPCR)	Low	Quantitative	Very High	Specialized equipment

How to measure gene expression?



Single cell experiments workflow



A comparative analysis of single-cell transcriptomic technologies in plants and animals

Vamsidhar Reddy Netla [a 1], Harshraj Shinde [b 1], Gulshan Kumar [b], Ambika Dudhate [c], Jong Chan Hong [d], Ulhas Sopanrao Kadam [d]

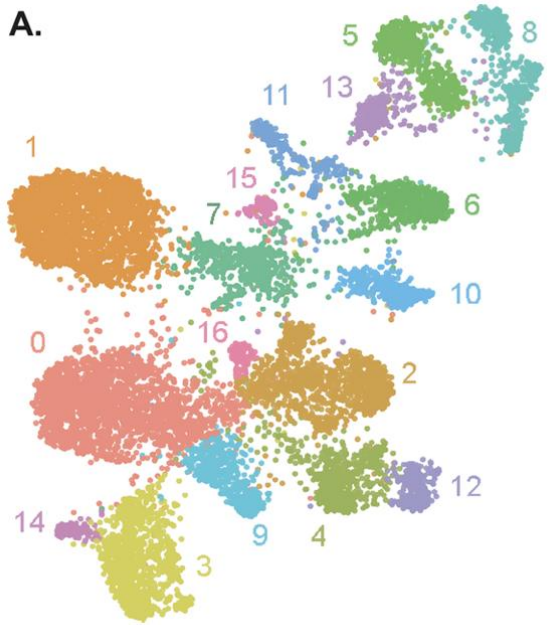
Show more
Add to Mendeley Share Cite
https://doi.org/10.1016/j.cpb.2023.100289
Get rights and content

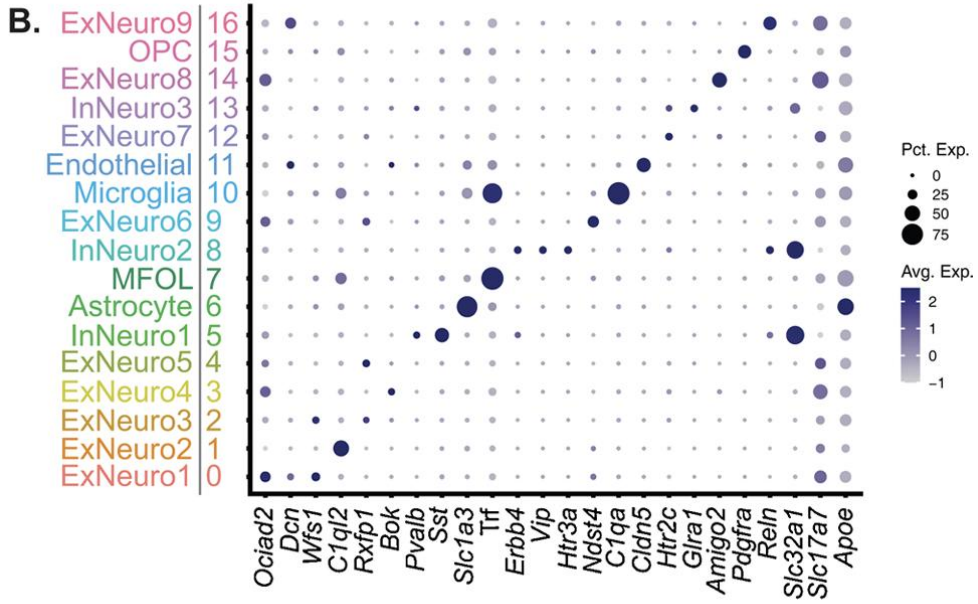
Examples of available data sets

Datasets (Showing 58 datasets)	Product	Species	Sample type	Cells or nuclei	Preservation
320k scFFPE From 8 Human Tissues 320k, 16-Plex	Flex Gene Expression v1.0	Human	Brain, Colon, Lung, Breast, Lymph node, Kidney, Skin, Cervix	N/A	Fixed
10k Mouse E18 Combined Cortex, Hippocampus and Subventricular Zone Cells, Chromium NextGEM Single Cell 3'	Universal 3' Gene Expression v3.1	Mouse	Brain	Cells	NA
10k Mouse E18 Combined Cortex, Hippocampus and Subventricular Zone Cells, Chromium GEM-X Single Cell 3'	Universal 3' Gene Expression v4	Mouse	Brain	Cells	NA
10k Adult Mouse Brain Nuclei Isolated with Chromium Nuclei Isolation Kit, Chromium NextGEM Single Cell 3'	Universal 3' Gene Expression v3.1	Mouse	Brain	Nuclei	Fresh Frozen
10k Adult Mouse Brain Nuclei Isolated with Chromium Nuclei Isolation Kit, Chromium GEM-X Single Cell 3'	Universal 3' Gene Expression v4	Mouse	Brain	Nuclei	Fresh Frozen
10k Mouse Forebrain FFPE Tissue Dissociated using gentleMACS Dissociator, Singleplex Sample (Next GEM)	Flex Gene Expression v1.0	Mouse	brain	Cells	FFPE
Mouse Brain Nuclei Isolated with Chromium Nuclei Isolation Kit, SaltyEZ Protocol, and 10x Complex Tissue DP (CT Sorted and CT Unsorted)	Epi Multiome ATAC + Gene Expression v1.0	Mouse	brain	Nuclei	Fresh Frozen
Mixture of Lung Cancer and Glioblastoma FFPE Tissues Dissociated Manually or using gentleMACS Dissociator, Multiplexed Samples, 4 Probe Barcodes (Next GEM)	Flex Gene Expression v1.0	Human	lung, brain	Cells	FFPE
5k Adult Mouse Brain Nuclei Isolated with Chromium Nuclei Isolation Kit	Universal 3' Gene Expression v3.1	Mouse	brain	Nuclei	Fresh Frozen
8k Adult Mouse Cortex Cells, ATAC v2, Chromium Controller	Epi ATAC v2	Mouse	brain	Nuclei	N/A
8k Adult Mouse Cortex Cells, ATAC v2, Chromium X	Epi ATAC v2	Mouse	brain	Nuclei	N/A
8k Adult Mouse Cortex Cells, ATAC v1.1, Chromium X	Epi ATAC v1.1	Mouse	brain	Nuclei	N/A
Multiomic Integration Neuroscience Application Note: Single Cell Multiome RNA + ATAC Alzheimer's Disease Mouse Model Brain Coronal Sections from One Hemisphere Over a Time Course	Epi Multiome ATAC + Gene Expression v1.0	Mouse	brain	Nuclei	Fresh Frozen, FFPE
Mouse E18 Combined Cortex, Hippocampus and Subventricular Zone Nuclei Multiplexed, 12 CMOs: 3'v3.1 Targeted, Custom Neuroscience Panel	Universal 3' Gene Expression v3.1	Mouse	brain, cortex, hippocampus, subventricular zone	Nuclei	N/A
Flash-Frozen Human Healthy Brain Tissue (3k)	Epi Multiome ATAC + Gene Expression v1.0	Human	brain, cerebellum	Nuclei	Frozen
Fresh Embryonic E18 Mouse Brain (5k)	Epi Multiome ATAC + Gene Expression v1.0	Mouse	cortex, hippocampus, ventricular zone, brain	Nuclei	Fresh

Dataset we will analyse today

- Alzheimer's disease model: ApoE knock out
- Single nucleus data
- 7 pooled hippocampi samples from 9-month-old C57BL/6 J wild type (WT) mice (ID GSM5067107)
- 7 pooled hippocampi samples from 9-month-old *Apoe* knockout (EKO) mice (ID GSM5067109)





differences in specific cell types.

All analyses were performed on GEO dataset GSE166261 [dataset] (Shi et al., 2021a, b) comprised of snRNAseq data generated by droplet-based Chromium Single Cell 3' Reagent Kit (10x Genomics) on 7 pooled hippocampi samples from 9-month-old C57BL/6 J wild type (WT) mice (Sample GSM5067107) and 7 pooled hippocampi samples from 9-month-old *Apoe* knockout (EKO) mice (Sample GSM5067109). The initial EKO mouse was generated using the 129P2/OlaHsd-derived E14Tg2a ES cell line (Piedrahita et al., 1992). The mice used to generate the samples for the snRNAseq data for this study were produced by backcrossing at least 10 generations to C57BL/6 J inbred mice.

Short Communication

Analysis of differential gene expression and transcript usage in hippocampus of *Apoe* null mutant mice: Implications for Alzheimer's disease

Andrew E. Weller*, Glenn A. Doyle, Benjamin C. Reiner, Richard C. Crist, Wade H. Berrettini

Center for Neurobiology and Behavior, Department of Psychiatry, Perelman School of Medicine, University of Pennsylvania, Philadelphia, PA, 19104, United States

General workflow

- Data import and Seurat object generation (already done)
- QC for mitochondrial genes, nr of genes and expression level
- Normalization: normalizing gene expression level for each cell
- Feature selection of variable genes
- Scaling: scaling expression for each gene to avoid overrepressentation of highly expressed genes
- Linear dimensional reduction: PCA
- Clustering: k-means for cluster selection
- Non-linear dimensional reduction: UMAP
- Feature selection and cluster annotation: looking at cluster specific markers
- (optional) Cell mapping, subclustering

How to find the data (example: GEO Gene Expression Omnibus)

ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSM5067109

Scope: Self Format: HTML Amount: Quick GEO accession: GSE166261 GO

Series GSE166261 Query DataSets for GSE166261

Status	Public on Feb 06, 2021
Title	Overexpressing low-density lipoprotein receptor reduces tau-associated neurodegeneration via apoE-dependent and independent mechanisms
Organism	Mus musculus
Experiment type	Expression profiling by high throughput sequencing

Samples (6)
Less...

GSM5067107	Isolated single nuclei from 7 pooled hippocampi of 9-month old WT mice
GSM5067108	Isolated single nuclei from 7 pooled hippocampi of 9-month old LDLR mice
GSM5067109	Isolated single nuclei from 7 pooled hippocampi of 9-month old EKO mice
GSM5067110	Isolated single nuclei from 8 pooled hippocampi of 9-month old P301S mice
GSM5067111	Isolated single nuclei from 7 pooled hippocampi of 9-month old P301S/LDLR mice
GSM5067112	Isolated single nuclei from 7 pooled hippocampi of 9-month old P301S/EKO mice

differences in specific cell types.

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Series (1) GSE166261 Overexpressing low-density lipoprotein receptor reduces tau-associated neurodegeneration via apoE-dependent and independent mechanisms

Relations

BioSample	SAMN17812771
SRA	SRX10038115

Supplementary file	Size	Download	File type/resource
GSM5067109_EKO_barcodes.tsv.gz	23.9 Kb	(ftp)(http)	TSV
GSM5067109_EKO_genes.tsv.gz	212.7 Kb	(ftp)(http)	TSV
GSM5067109_EKO_matrix.mtx.gz	14.7 Mb	(ftp)(http)	MTX

SRA Run Selector

Raw data are available in SRA

Processed data provided as supplementary file

Download all 3 files

Introduction into Seurat



satijalab.org/seurat/

Seurat 5.2.0 | Install | Get started | Vignettes | Extensions | FAQ | News



Introductory Vignettes

PBMC 3K guided tutorial

Data visualization vignette

SCTransform, v2 regularization

Using Seurat with multi-modal data

Seurat v5 Command Cheat Sheet

Data Integration

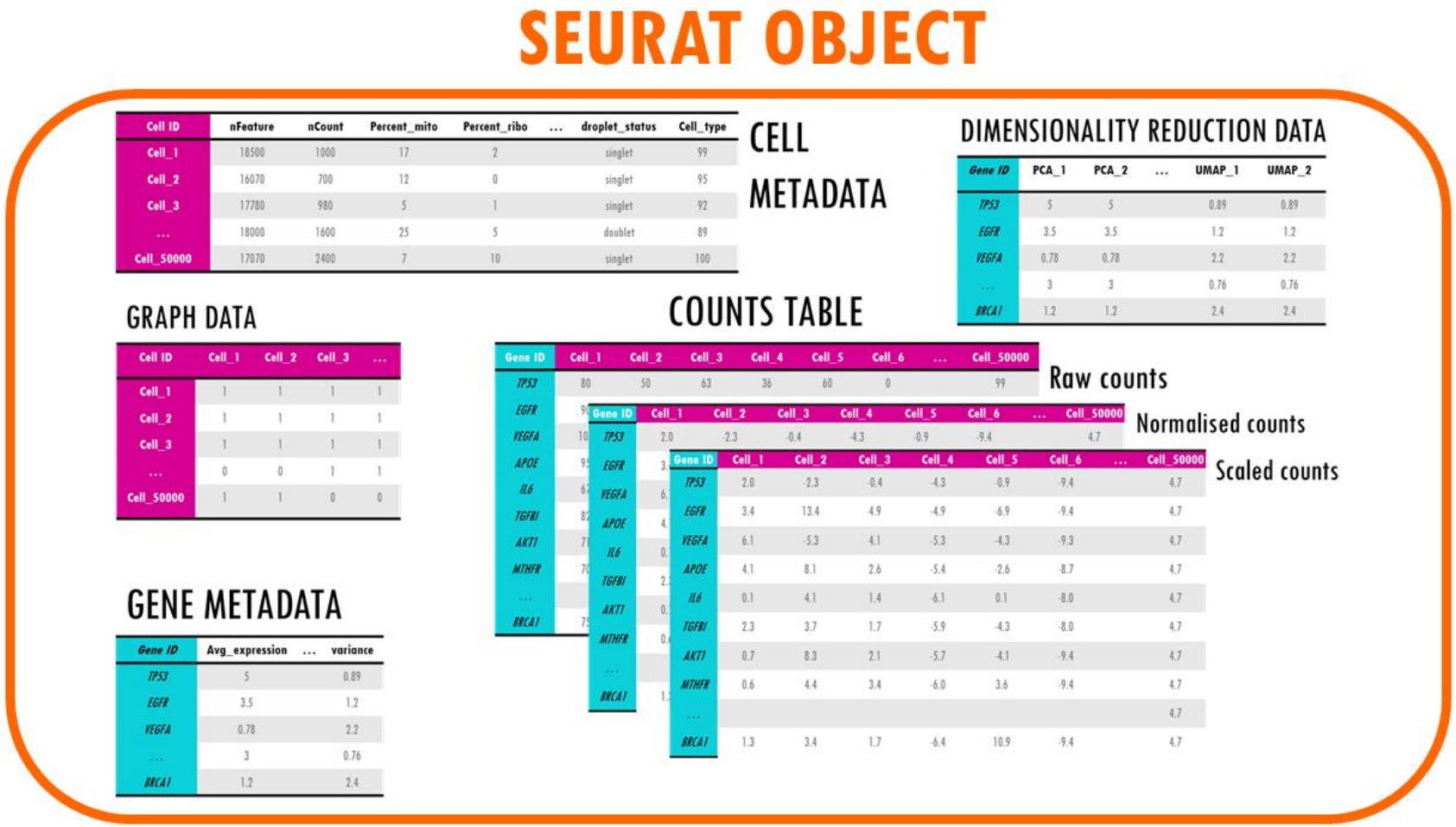
Introduction to scRNA-seq integration

Seurat v5

https://satijalab.org/seurat/

Introduction into Seurat

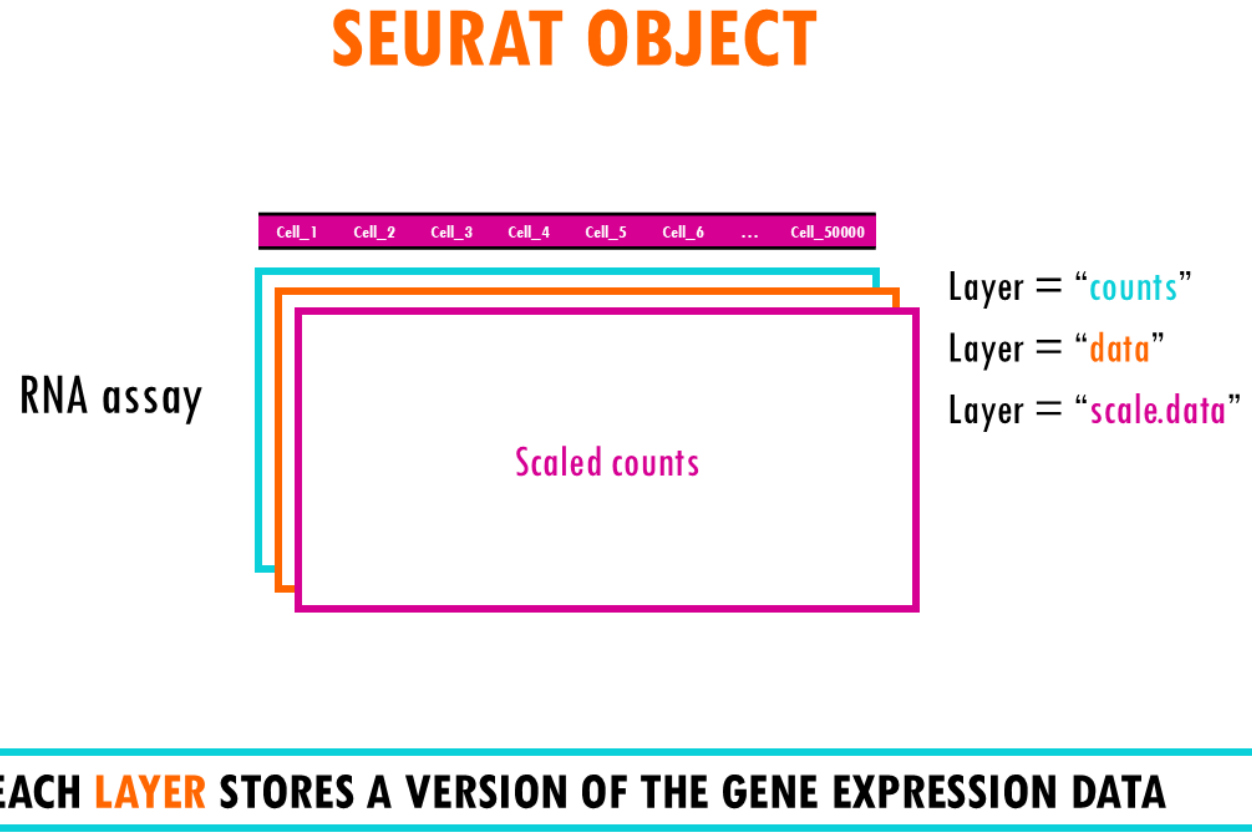
- What it is
- Why we use it



https://biostatsquid.com/seurat-objects-explained/

Introduction into Seurat

- What it is
- Why we use it



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General workflow

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- Non-linear dimensional reduction: UMAP
- Feature selection and cluster annotation: looking at cluster specific markers
- (optional) Cell mapping, subclustering

Part 1: Data import

```
library(dplyr)
library(Seurat)
library(patchwork)

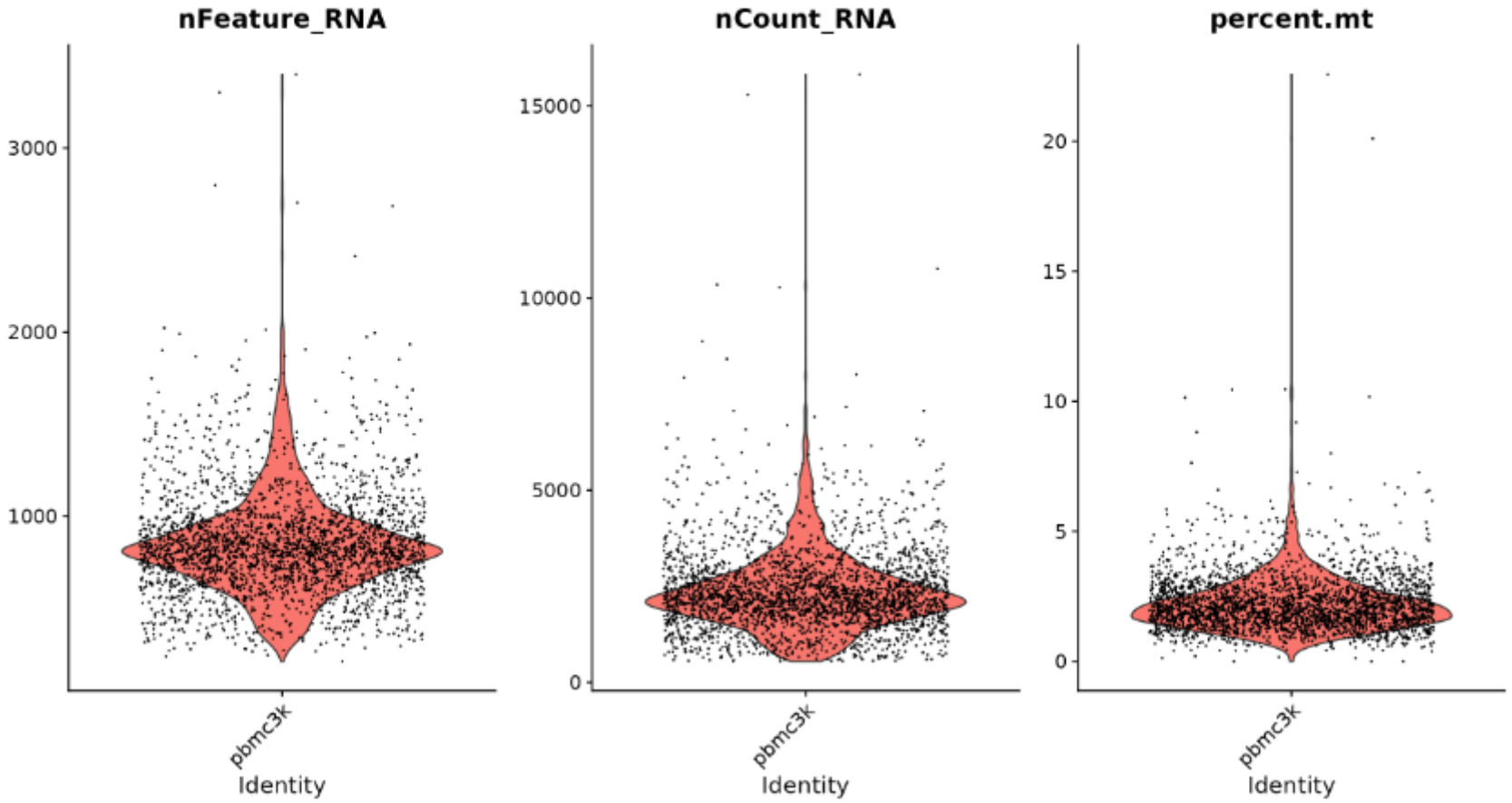
# Load the PBMC dataset
pbmc.data <- Read10X(data.dir = "/brahms/mollag/practice/filtered_gene_bc_matrices/hg19/")
# Initialize the Seurat object with the raw (non-normalized data).
pbmc <- CreateSeuratObject(counts = pbmc.data, project = "pbmc3k", min.cells = 3, min.features = 200)
pbmc
```

```
## An object of class Seurat
## 13714 features across 2700 samples within 1 assay
## Active assay: RNA (13714 features, 0 variable features)
##  1 layer present: counts
```

https://satijalab.org/seurat/articles/pbmc3k_tutorial

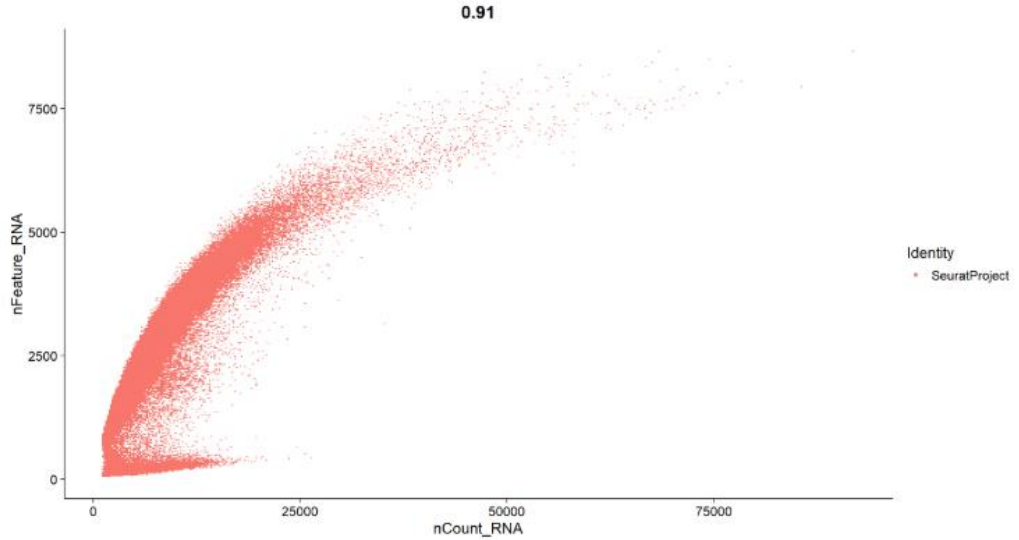
Part 2: Data filter

```
# Visualize QC metrics as a violin plot
VlnPlot(pbmc, features = c("nFeature_RNA", "nCount_RNA", "percent.mt"), ncol = 3)
```

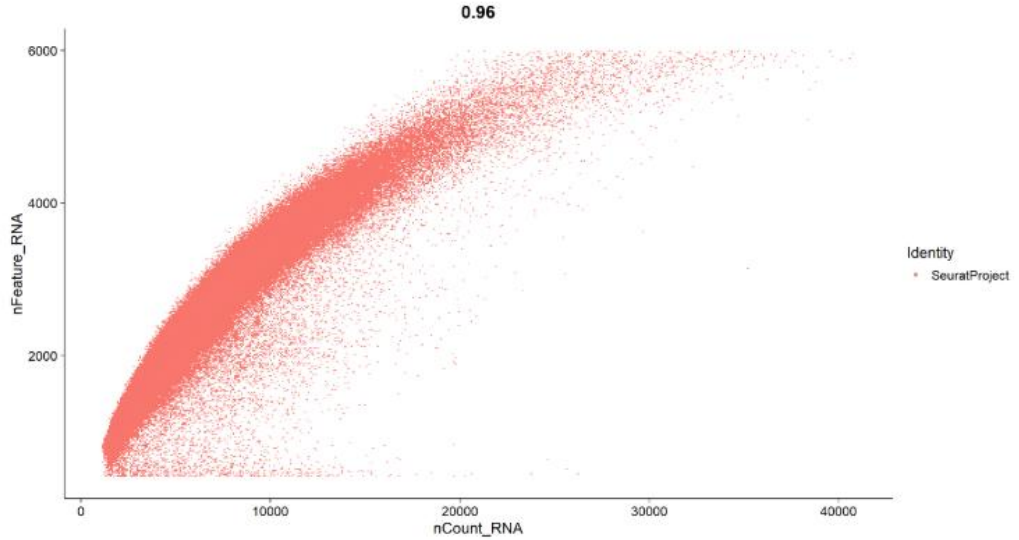


Part 2: Data filter

```
# FeatureScatter is typically used to visualize feature-feature relationships, but can be used for anything calculated by th
e object, i.e. columns in object metadata, PC scores etc.
FeatureScatter(obj, feature1 = "nCount_RNA", feature2 = "nFeature_RNA")
```

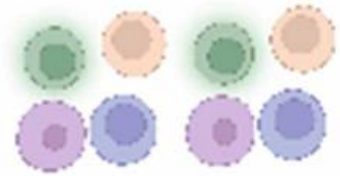


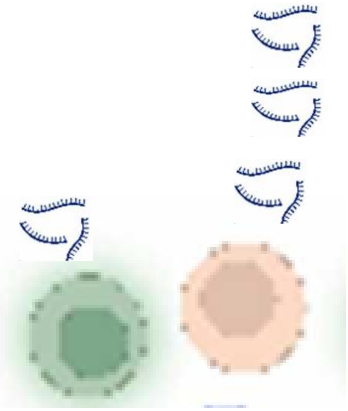

```
obj <- subset(obj, subset = nFeature_RNA > 400 & nFeature_RNA < 6000)
FeatureScatter(obj, feature1 = "nCount_RNA", feature2 = "nFeature_RNA")
```



https://biostatsquid.com/seurat-objects-explained/

Normalization







	Cell 1	Cell 2	Cell 3
Total reads	3	300	100

RAW

Gene A	1	100	50
Gene B	2	200	50

LOG2 NORMALISATION

NORMALISED

Gene A	8.11	8.11	8.52
Gene B	8.81	8.81	8.52

Gene X cell Y: $\log_e(Val_X/Total_Y*10000 + 1)$

Gene A cell 1: $\log_e(1/3*10000 + 1)$

Gene A cell 3: $\log_e(50/100*10000 + 1)$

Normalization: This step addresses differences between cells by adjusting for sequencing depth or other technical variations, ensuring that gene expression values are comparable across cells.

Part 2: Data normalization

Normalizing the data

After removing unwanted cells from the dataset, the next step is to normalize the data. By default, we employ a global-scaling normalization method "LogNormalize" that normalizes the feature expression measurements for each cell by the total expression, multiplies this by a scale factor (10,000 by default), and log-transforms the result. In Seurat v5, Normalized values are stored in `obj[["RNA"]]$data`.

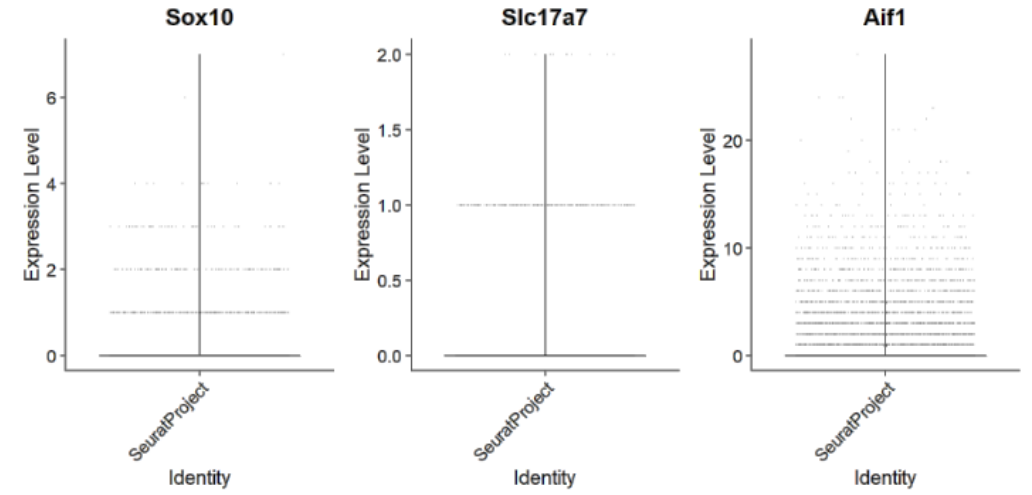
```
#obj <- NormalizeData(obj, normalization.method = "LogNormalize", scale.factor = 1e4)
```

For clarity, in this previous line of code (and in future commands), we provide the default values for certain parameters in the function call. However, this isn't required and the same behavior can be achieved with:

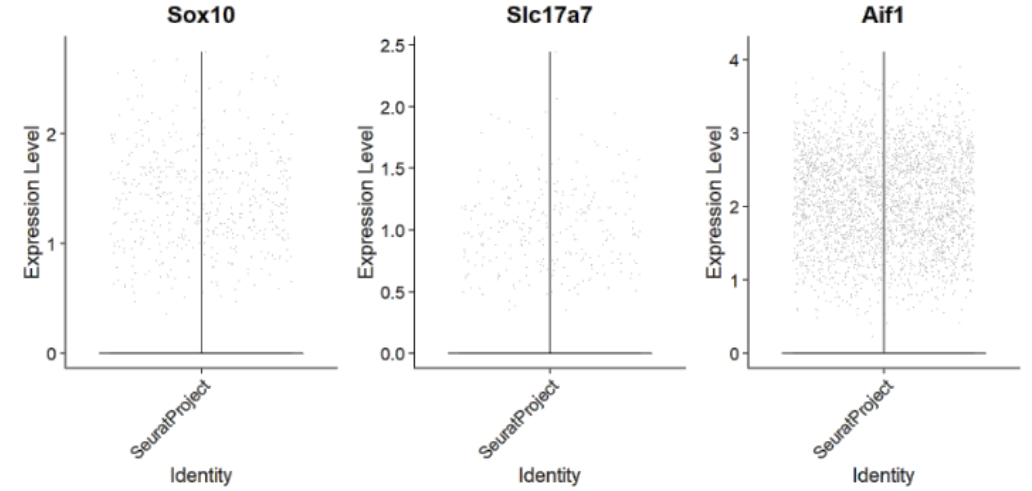
```
obj <- NormalizeData(obj)
```

Compare Normalization with Counts

```
VlnPlot(obj, features = c("Sox10", "Slc17a7", "Aif1"), ncol = 3, layer = "counts", alpha = 0.1)
```



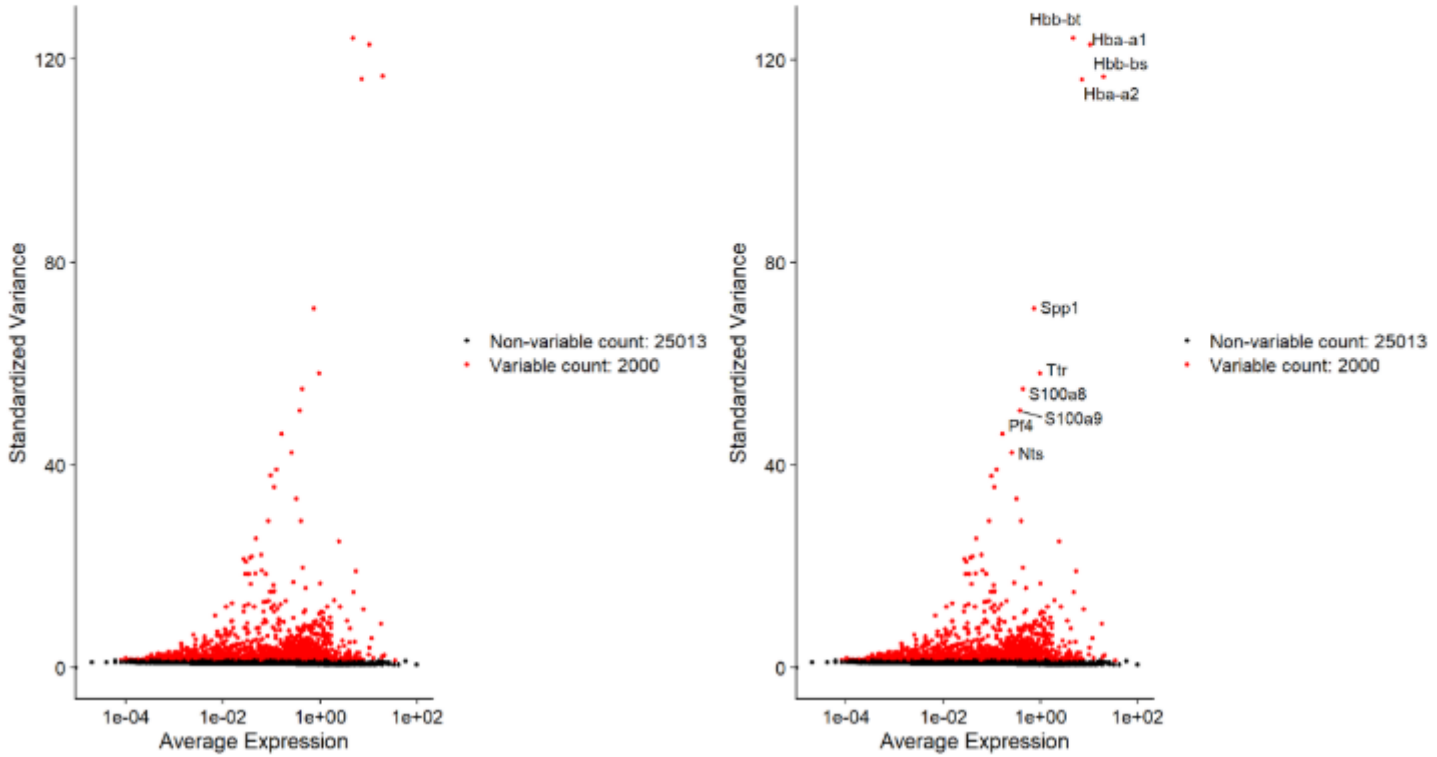

```
# We then visualize normalized data
VlnPlot(obj, features = c("Sox10", "Slc17a7", "Aif1"), ncol = 3, layer = "data", alpha = 0.1)
```



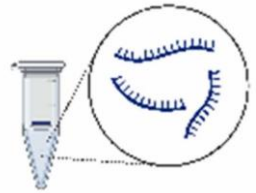
Part 3: Find variable features (feature selection)

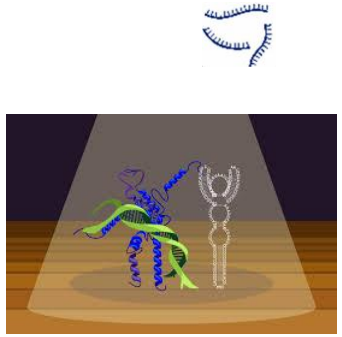
```
obj <- FindVariableFeatures(obj)
# Identify the 10 most highly variable genes
top10 <- head(VariableFeatures(obj), 10)

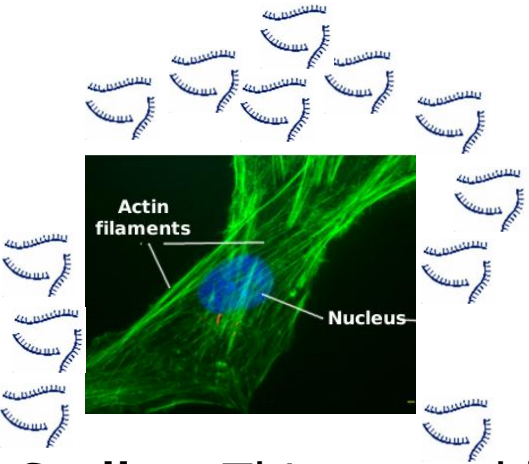
# plot variable features with and without labels
plot1 <- VariableFeaturePlot(obj)
plot2 <- LabelPoints(plot = plot1, points = top10, repel = TRUE)
plot1 + plot2
```



Scaling







RAW

	Cell 1	Cell 2	Cell 3
Gene A	1	100	50
Gene B	2	200	50

LOG2 NORMALISATION

NORMALISED

Gene A	8.11	8.11	8.52
Gene B	8.81	8.81	8.52

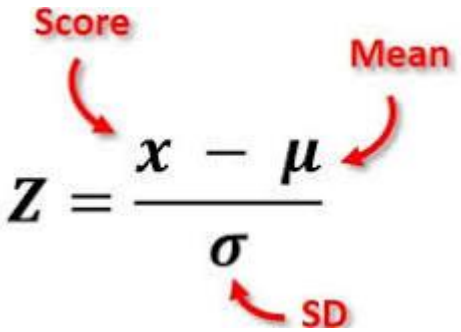
Z-SCORE

SCALED

Gene A	-0.57	-0.57	+1.19
Gene B	+0.58	+0.58	-1.16

Scaling: This step addresses differences between genes by adjusting for varying ranges in expression levels across genes, making sure that no single gene with very high expression dominates the analysis.

Part 4: scale the data



Scaling the data

Next, we apply a linear transformation ('scaling') that is a standard pre-processing step prior to dimensional reduction techniques like PCA. The `ScaleData()` function:

- Shifts the expression of each gene, so that the mean expression across cells is 0
- Scales the expression of each gene, so that the variance across cells is 1
  - This step gives equal weight in downstream analyses, so that highly-expressed genes do not dominate
- The results of this are stored in `obj[["sketch"]]$scale.data`
- By default, only variable features are scaled.
- You can specify the `features` argument to scale additional features

```
obj <- ScaleData(obj)
```

Part 5: PCA

For the first principal components, Seurat outputs a list of genes with the most positive and negative loadings, representing modules of genes that exhibit either correlation (or anti-correlation) across single-cells in the dataset.

```
obj <- RunPCA(obj, features = VariableFeatures(object = obj))
```

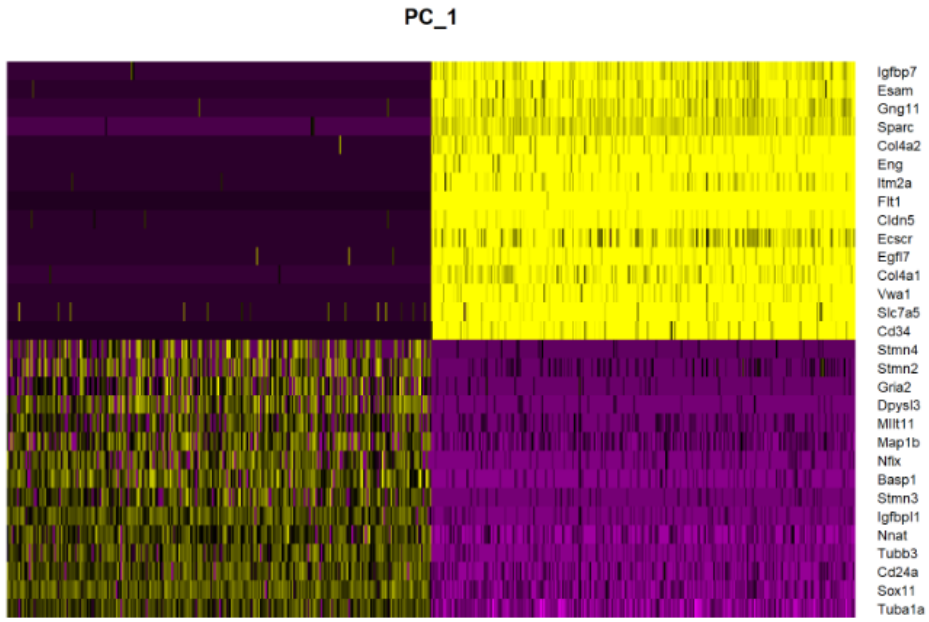
Seurat provides several useful ways of visualizing both cells and features that define the PCA, including `VizDimReduction()`, `DimPlot()`, and `DimHeatmap()`

```
# Examine and visualize PCA results a few different ways
print(obj[['pca']], dims = 1:5, nfeatures = 5)
```

```
## PC_ 1
## Positive:  Tuba1a, Sox11, Cd24a, Tubb3, Nnat
## Negative:  Igfbp7, Esam, Gng11, Sparc, Col4a2
## PC_ 2
## Positive:  C1qc, C1qb, Tyrobp, C1qa, Fcer1g
## Negative:  Tmsb10, Tuba1a, Tsc22d1, Sparcl1, Serpinh1
## PC_ 3
## Positive:  Tubb3, Tmsb10, Mllt11, Stmn2, Stmn3
## Negative:  Hmgb2, Dbi, Phgdh, Fabp7, Cks2
## PC_ 4
## Positive:  Vtn, Ndufa4l2, Kcnj8, Higd1b, Abcc9
## Negative:  Cldn5, Vwa1, Cd34, Pglyrp1, Ctla2a
## PC_ 5
## Positive:  Birc5, Ube2c, Ccna2, Nusap1, Spc25
## Negative:  Cimap3, Clu, Foxj1, Pierce1, Rsph1
```

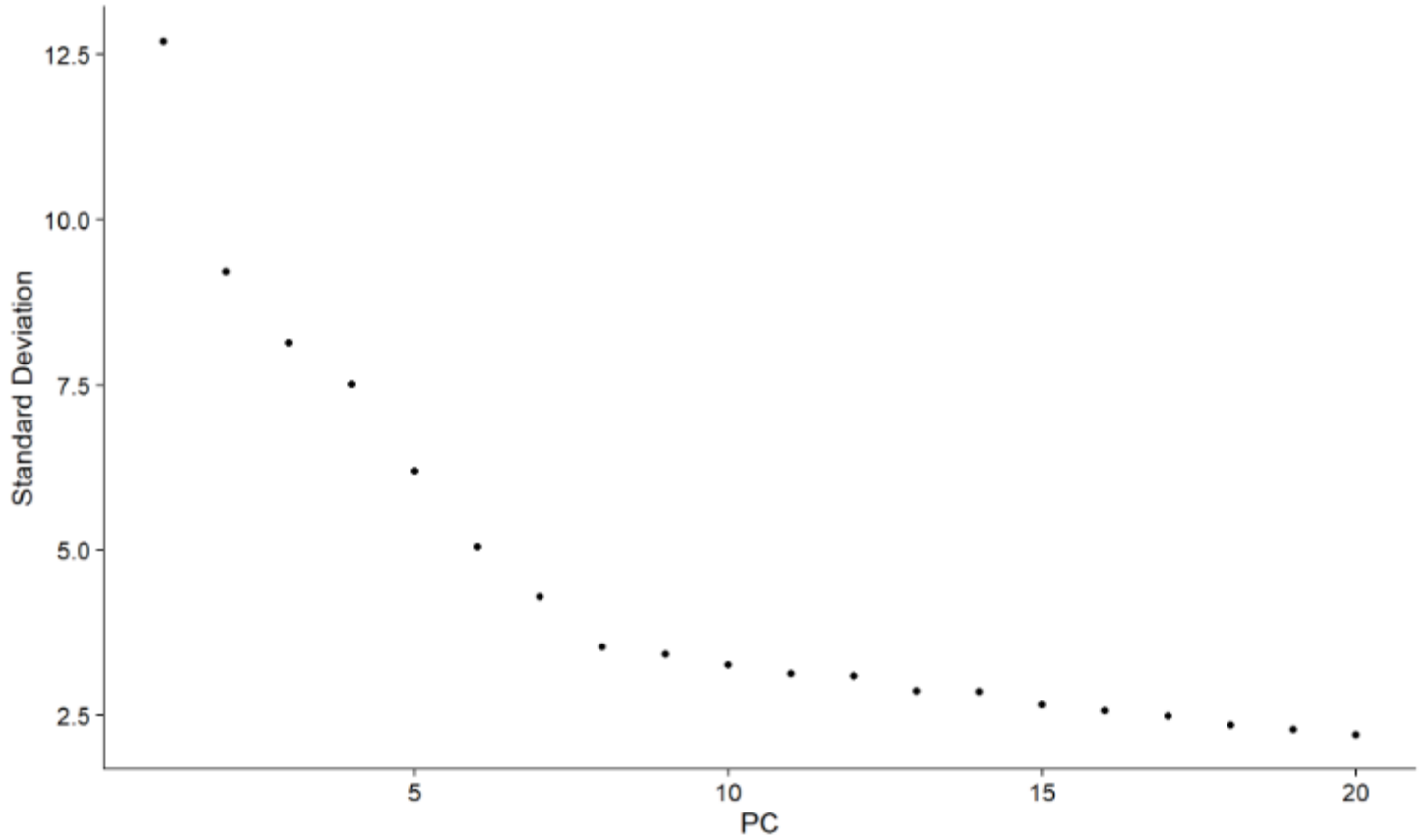
```
VizDimLoadings(obj, dims = 1:2, reduction = 'pca')
```

```
DimHeatmap(obj, dims = 1, cells = 500, balanced = TRUE)
```



Part 5: PCA

```
ElbowPlot(obj)
```



Part 6: Clustering

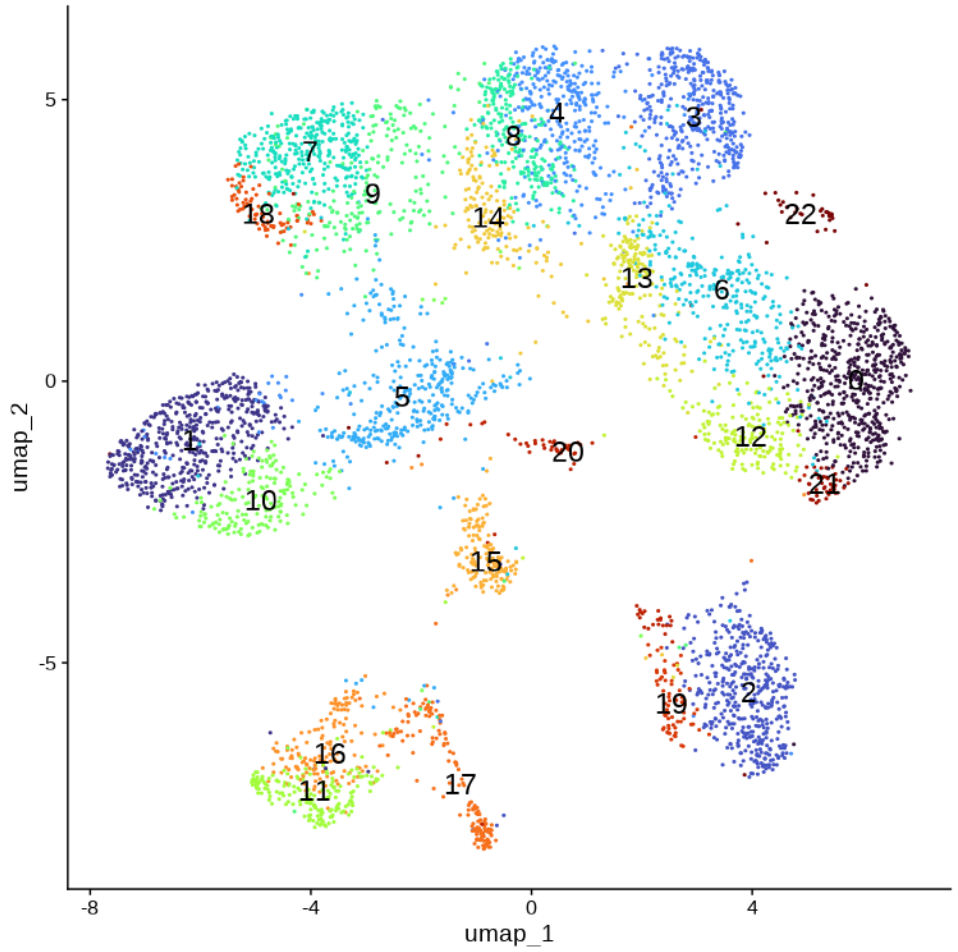
```
obj <- FindNeighbors(obj,reduction = "pca", dims = 1:50)
obj <- FindClusters(obj, resolution = 2) # the higher the number the higher the clusters
```

```
## Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck
##
## Number of nodes: 49451
## Number of edges: 1969041
##
## Running Louvain algorithm...
## Maximum modularity in 10 random starts: 0.8779
## Number of communities: 54
## Elapsed time: 13 seconds
```

```
# Look at cluster IDs of the first 5 cells
head(Idents(obj), 5)
```

```
## AAACCTGAGATAGGAG-1 AAACCTGAGCGGCTTC-1 AAACCTGAGGAATCGC-1 AAACCTGAGGACACCA-1
##               <NA>               <NA>               <NA>               <NA>
## AAACCTGAGGCCCGTT-1
##                 17
## 54 Levels: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 ... 53
```

Part 7: Non-linear dimensional reduction (UMAP)



Day 4

Clustering of cells in Seurat

DEG and its interpretation

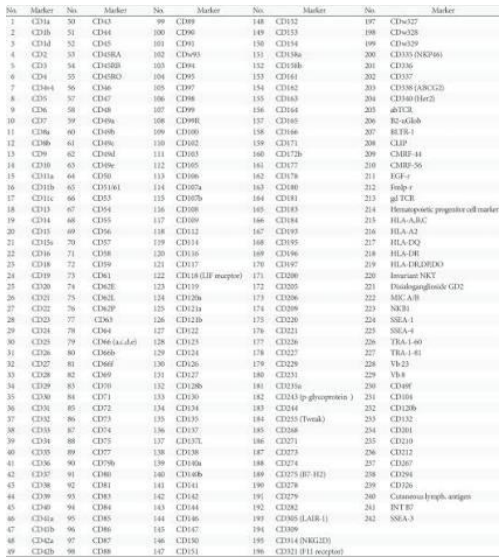
General workflow

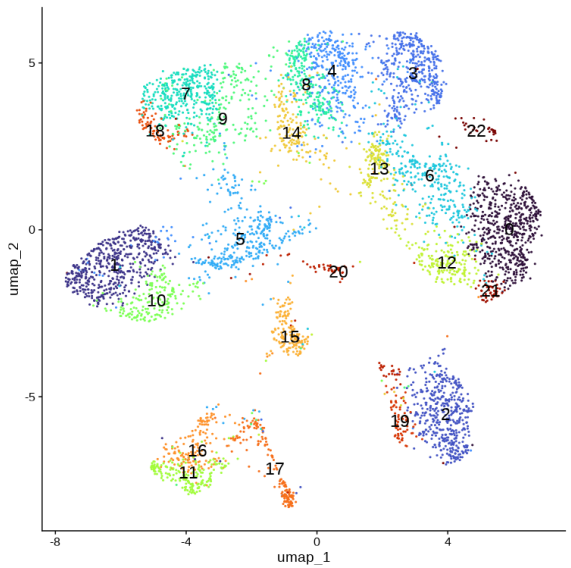
- Data import and Seurat object generation (already done)
- QC for mitochondrial genes, nr of genes and expression level
- Normalization: normalizing gene expression level for each cell
- Feature selection of variable genes
- Scaling: scaling expression for each gene to avoid overrepressentation of highly expressed genes
- Linear dimensional reduction: PCA
- Clustering: k-means for cluster selection
- Non-linear dimensional reduction: UMAP
- Feature selection and cluster annotation: looking at cluster specific markers
- (optional) Cell mapping, subclustering

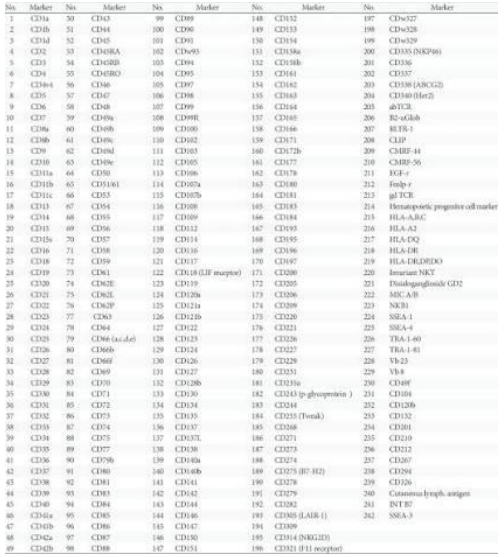
Cluster annotation

Check known markers against expression in identified clusters

OR: Extract markers from detected clusters and compare to published datasets

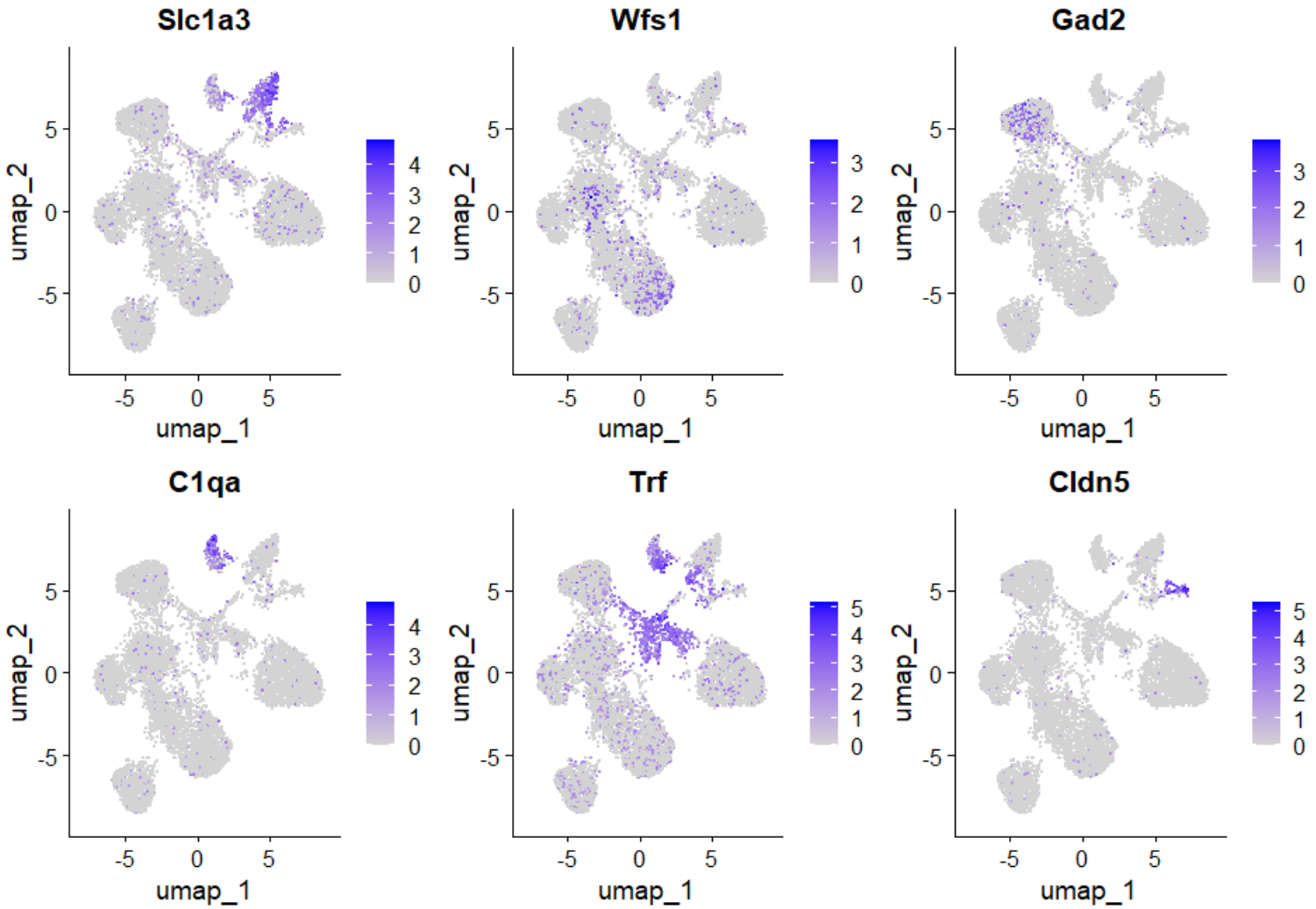






Part 8: Find markers

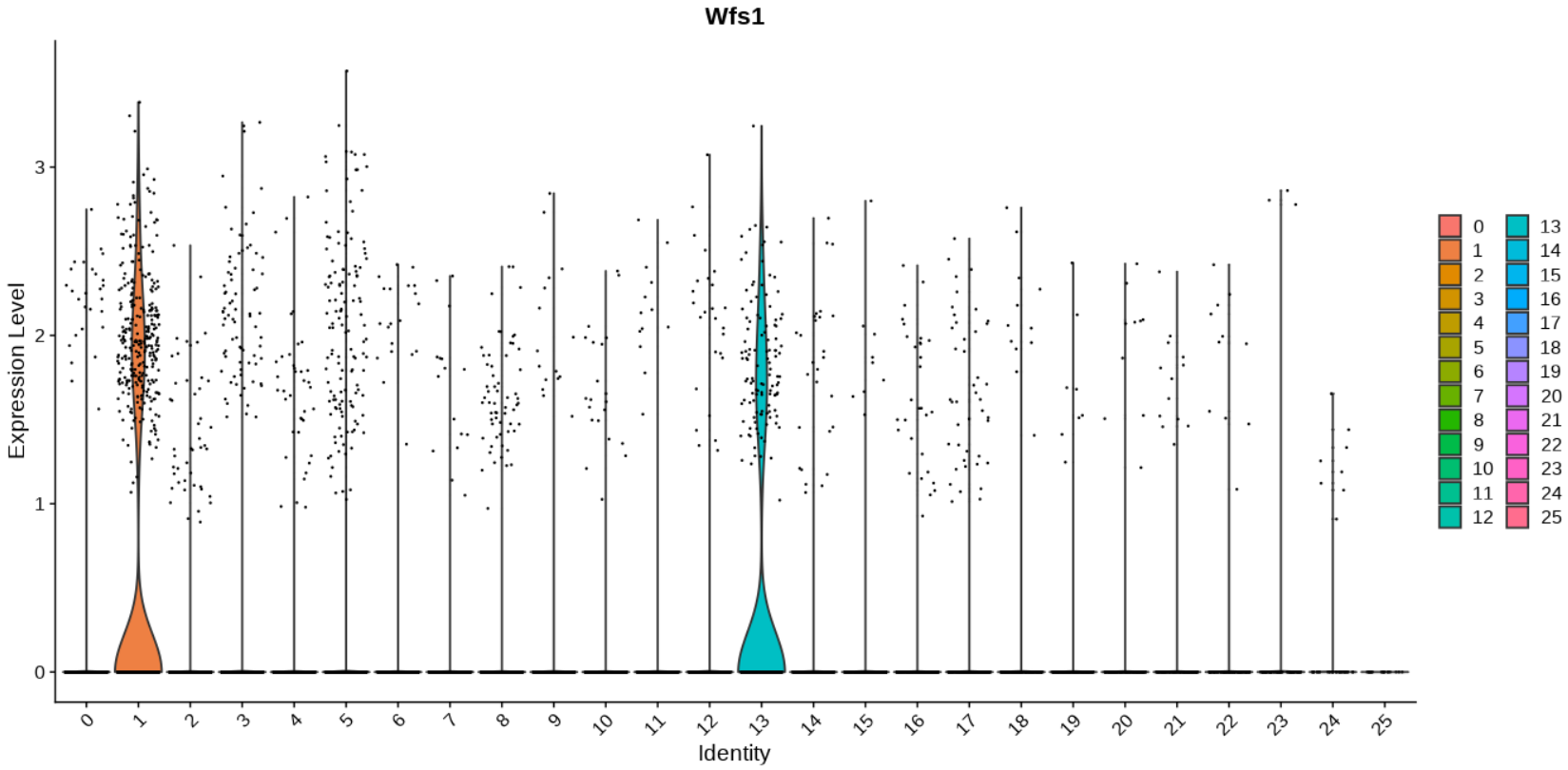
```r
```{r,fig.height = 7, fig.width = 10}
FeaturePlot(
  object = Seurat_AD_WT,
  features = c(
    "Slc1a3", #Astrocytes
    "Wfs1", # Glut Neurons or Slc17a7
    "Gad2", # GABA neurons
    "C1qa", # Microglia
    "Trf", # Myelin-forming mature oligodendrocytes
    "Cldn5"), # Endothelial cells
  ncol = 3)
```
```



Part 8: Find markers

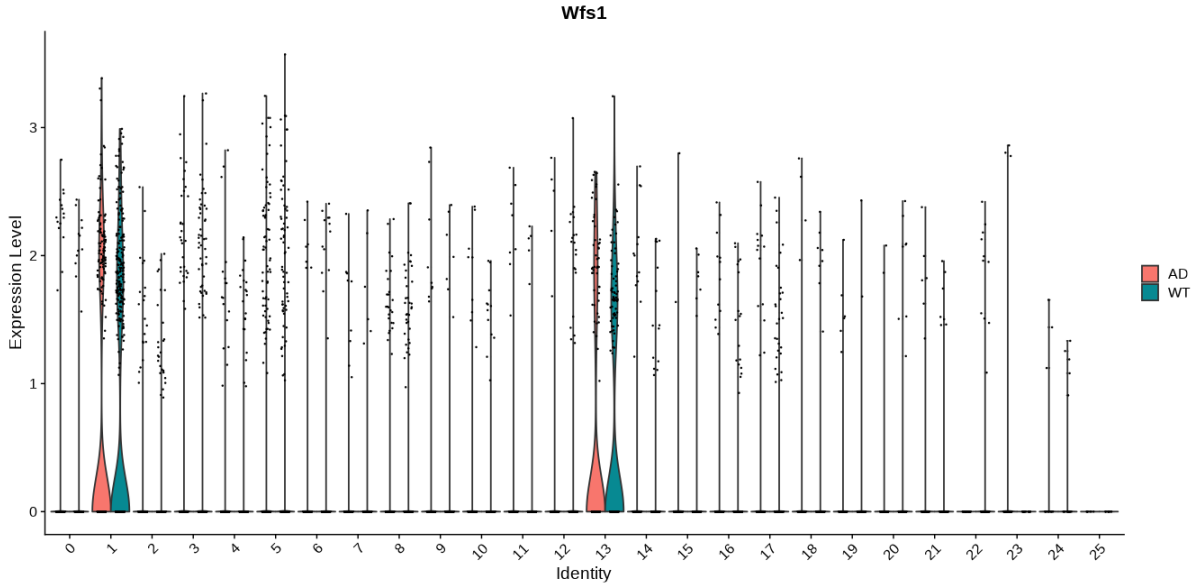
Differential gene expression

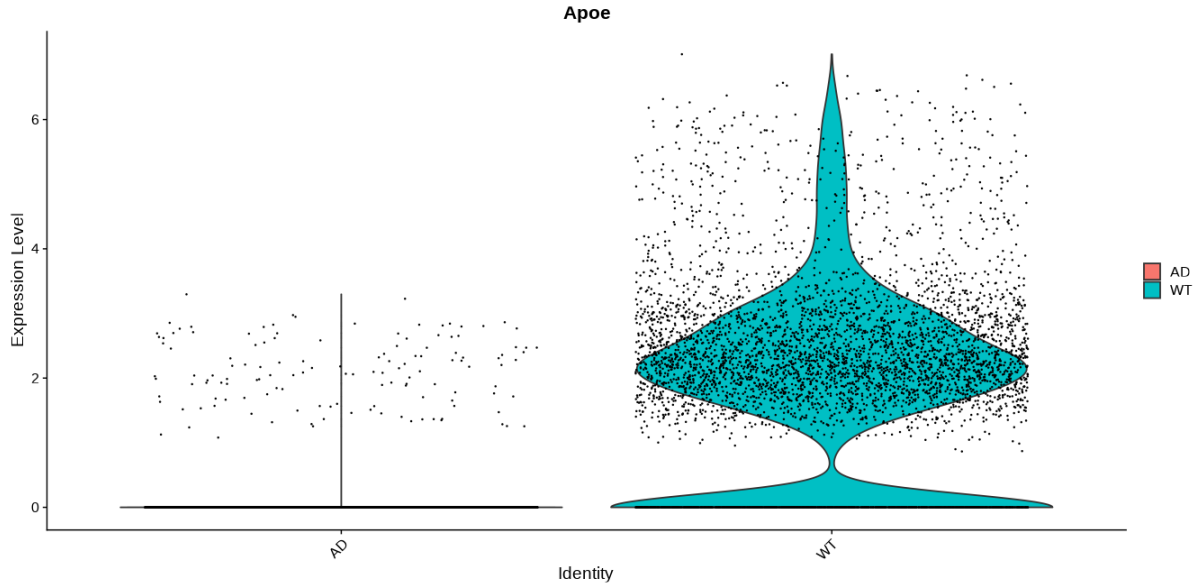
```
VlnPlot(obj, 'Dlx2')
```



Part 8: Find markers

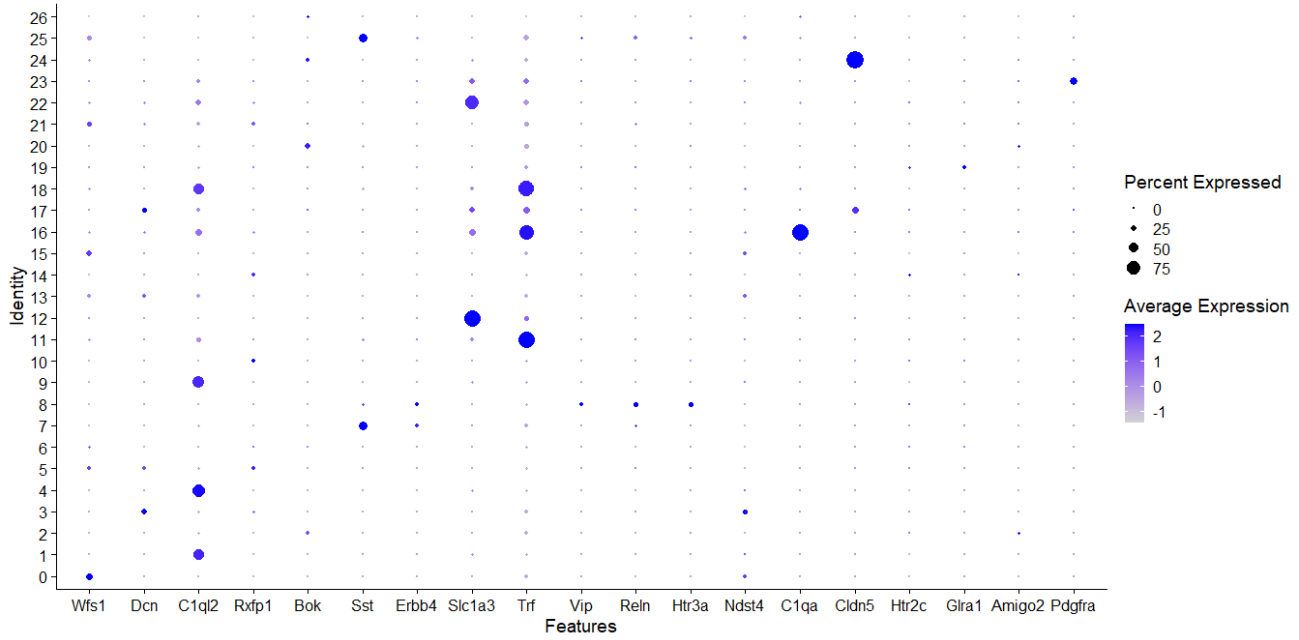
```r
```{r, fig.height = 7, fig.width = 14}
VlnPlot(Seurat_AD_WT, 'Wfs1')
VlnPlot(Seurat_AD_WT, 'Wfs1', split.by = 'condition')
VlnPlot(Seurat_AD_WT, 'Apoe', group.by = 'condition') # check knockout
```
```





Part 8: Find markers

```r
```{r, fig.height = 7, fig.width = 14}
DotPlot(Seurat_AD_WT, features= all_genes)
```


```

Part 8: Find markers

```r
```{r, fig.height = 7, fig.width = 14}
# find all markers of cluster 5
Idents(Seurat_AD_WT) <- "seurat_clusters"
cluster13.markers <- FindMarkers(Seurat_AD_WT, ident.1 = "13", only.pos = TRUE)
head(cluster13.markers, n = 10)
```

Markers

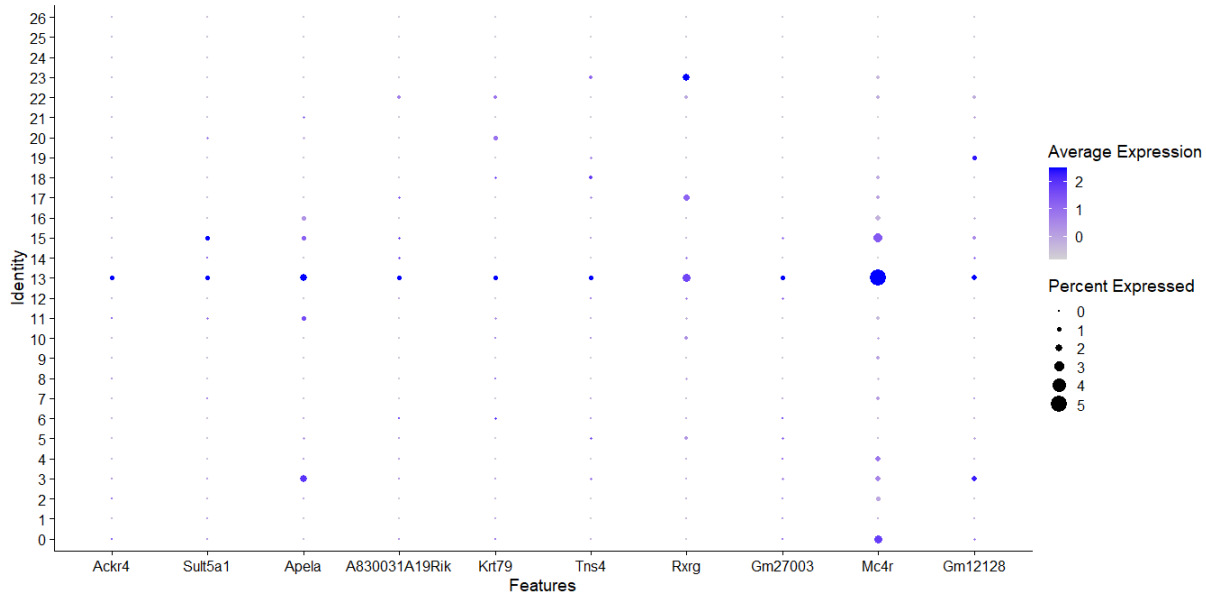
	p_val	avg_log2FC	pct.1	pct.2	p_val_adj
Cpne7	3.795059e-113	1.6148848	0.939	0.440	1.062541e-108
Pex5l	6.549430e-80	1.3841500	0.867	0.414	1.833709e-75
Man1a	1.168222e-55	1.7415038	0.408	0.123	3.270787e-51
Grin2b	6.088627e-55	0.6456170	1.000	0.913	1.704694e-50
Kcnn2	1.985708e-46	1.3211142	0.589	0.252	5.559586e-42
3110035E14Rik	1.610460e-42	1.1141428	0.614	0.279	4.508965e-38
Ntm	1.879055e-40	0.7167477	0.956	0.618	5.260977e-36
Chrd	8.107074e-40	1.0820214	0.683	0.354	2.269819e-35
Fam19a1	3.278474e-37	0.6712845	0.917	0.568	9.179072e-33
Fibcd1	4.322958e-37	1.6778961	0.267	0.076	1.210342e-32
Zeb2	6.237067e-33	0.7754559	0.853	0.579	1.746254e-28
Arhgef28	8.172101e-31	1.3445675	0.294	0.099	2.288025e-26
Mpped1	4.221407e-29	1.0897637	0.481	0.220	1.181910e-24
Brd9	7.888922e-28	0.7317513	0.836	0.619	2.208740e-23
Cadps2	6.051534e-27	1.3831535	0.256	0.086	1.694308e-22
Atp2b1	1.206030e-26	0.5222782	0.939	0.733	3.376643e-22
Grm5	2.330329e-26	0.5810710	0.939	0.721	6.524456e-22

- **avg_logFC**: log fold-change of the average expression between the two groups. Positive values indicate that the gene is more highly expressed in the first group
- **pct.1**: The percentage of cells where the gene is detected in the first group
- **pct.2**: The percentage of cells where the gene is detected in the second groupp_val_adj: Adjusted p-value, based on bonferroni correction using all genes in the dataset
```

Part 8: Find markers

```
### Plot the markers
```{r, fig.height = 7, fig.width = 14}
cluster13.markers %>%
    dplyr::filter(avg_log2FC > 2 & p_val < 0.05) %>%
    slice_max(avg_log2FC, n = 10) -> top10

Idents(Seurat_AD_WT) <- "seurat_clusters"
DotPlot(Seurat_AD_WT, features = rownames(top10))
DotPlot(Seurat_AD_WT, features = rownames(top10), group.by = "condition")
```
```



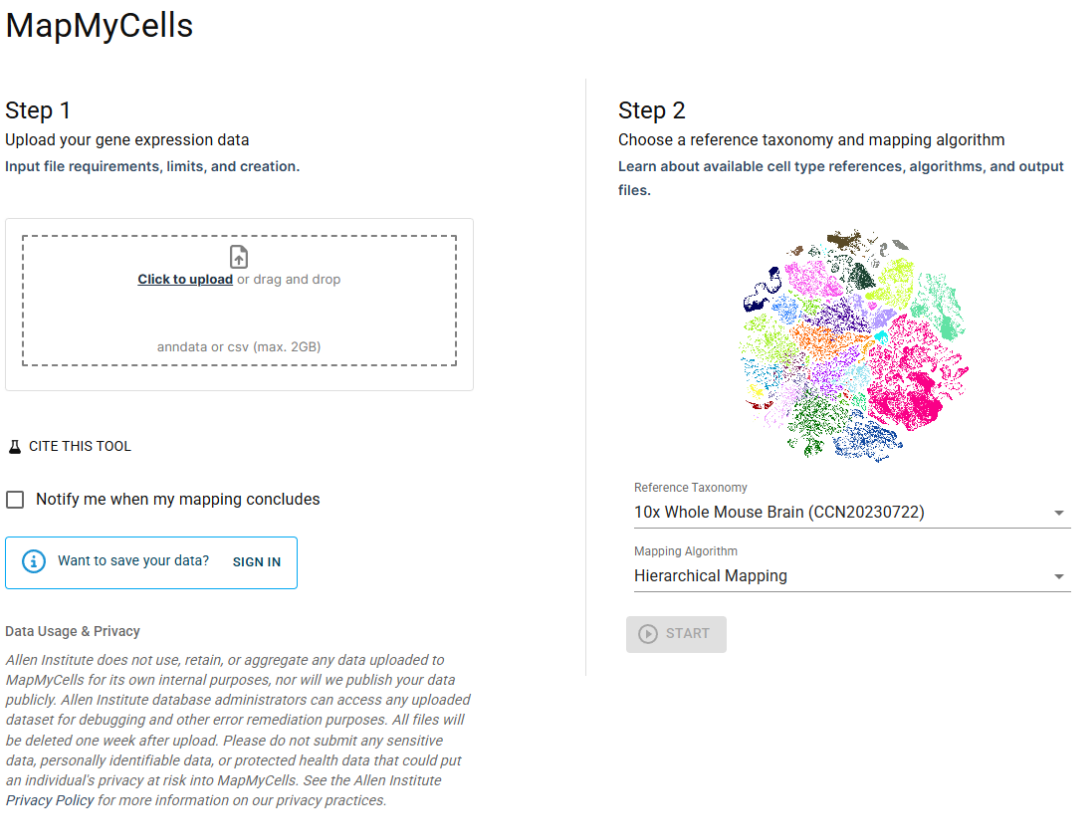
Part 8: Find markers

Aspect	FindMarkers(ident.1 = 5)	FindMarkers(ident.1 = "5_WT", ident.2 = "5_KO")	FindConservedMarkers(ident.1 = 5, grouping.var = "condition")
Purpose	Find genes that define cluster 5 identity	Find genes differentially expressed between conditions in cluster 5	Find genes that consistently mark cluster 5 regardless of condition
Comparison	Cluster 5 vs. all other cells	WT cluster 5 vs. KO cluster 5	Within-condition cluster-vs-cluster comparisons, then combined
Output significance	Single p-value per gene	Single p-value per gene	Separate p-values per condition + meta-analysis p-value
Biological question	What makes cluster 5 unique?	Which genes respond differently to genotype in cluster 5?	Which genes robustly identify cluster 5 across both genotypes?

MapMyCells

```
# Export data to MapMyCells
We will export the count matrix and map the cells with Allen Brain atlas
see: https://portal.brain-map.org/atlases-and-data/bkp/mapmycells
use: https://knowledge.brain-map.org/mapmycells/process/
## Extract the Count Matrix

```{r,fig.height = 7, fig.width = 10}
mat <- GetAssayData(object = subset(Seurat_AD_WT, downsample = 1000, subset = seurat_clusters %in% c(13)), assay = "RNA", slot = "counts")
# we need to transpose the matrix, as MapMyCells expects matrix in which rows are "cells" and columns are genes!
mat_t <- t(mat)
write.csv(mat_t, "./output/count_matrix_cluster13.csv")
```
```



MapMyCells

MOUSE

Whole Mouse Brain Cell Type Taxonomy (AIT21)

In this documentation:

- Overview
- Resources
- Download the Data
- Publication
- ABC Atlas
- MapMyCells

Overview

Using advanced technologies that profile individual cells, we unveiled an atlas cataloging the location and type of every cell in the adult mouse brain. From >8 million cells in multiple data modalities, the team identified over 5,300 cell types – far more than known before – and pinpointed their locations within the brain's intricate geography. This study is part of a set of 10 papers in Nature that will likely define the mouse whole brain for years to come.

Taxonomy Name: 20230722_BICCN-Adult_Mouse_WholeBrain_10X

Taxonomy Description: Whole Mouse Brain Taxonomy

Species: Mouse

CCN Number: CCN20230722

AIT Number: 21

Resources

Download the Data

Download and explore the data via Python notebook

Publication

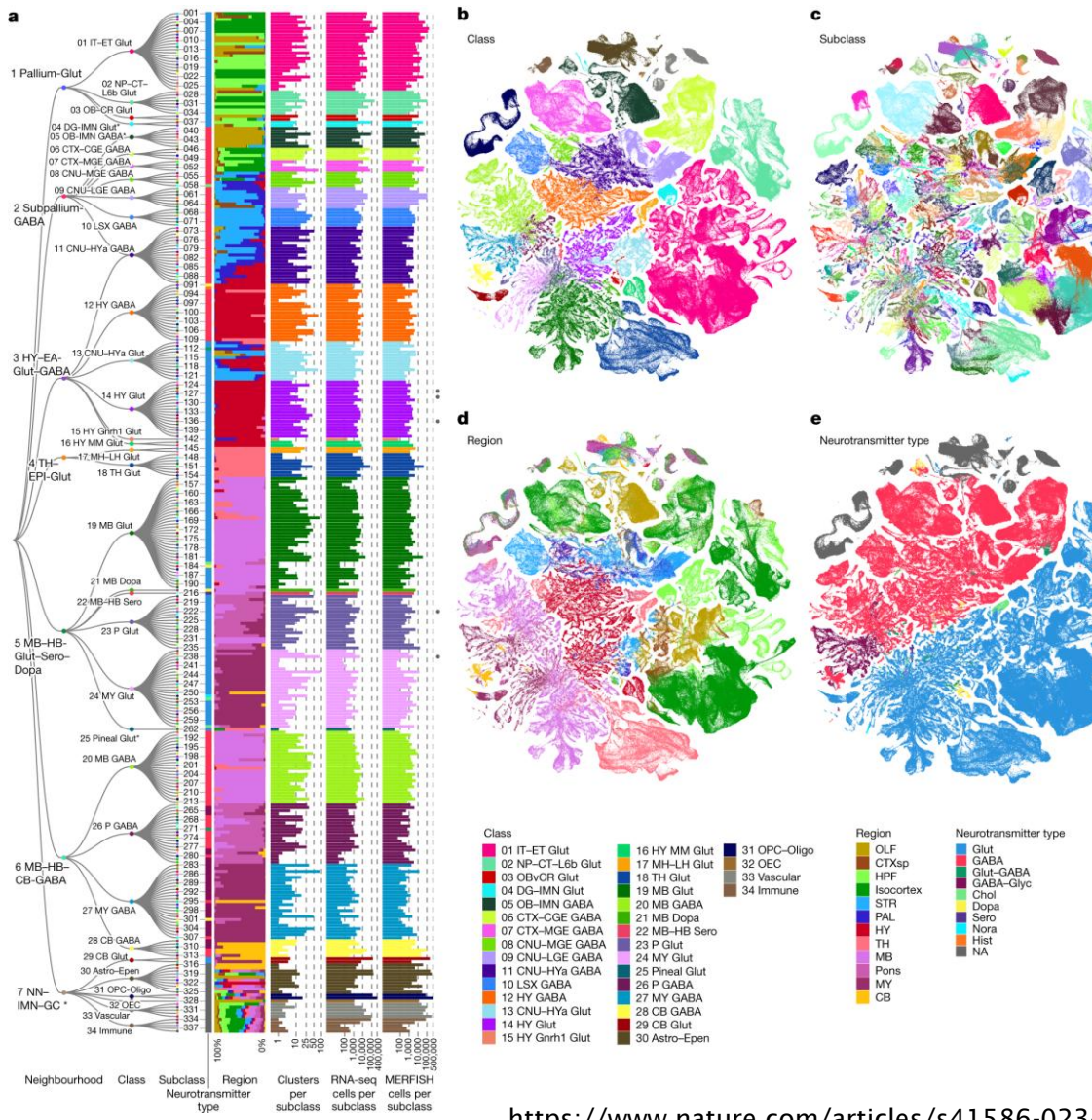
A high-resolution transcriptomic and spatial atlas of cell types in the whole mouse brain

ABC Atlas

Explore the data in the ABC Atlas

MapMyCells

Map your data using the whole mouse brain taxonomy as a reference



https://www.nature.com/articles/s41586-023-06812-z

Thank you for the attention

Please evaluate our course!



Lukasz



Kaja